IBM Data Analyst Capstone Project Report

Stack Overflow Developer Survey Analysis

Insights into Programming Languages, Databases, Platforms, and Developer Demographics

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03.03.2026

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EXECUTIVE SUMMARY

The Stack Overflow Developer Survey dataset contains responses from approximately 90,000 developers worldwide, providing insights into programming languages, platforms, databases, and developer characteristics. For this analysis, a randomized subset of the dataset was used to explore technology trends and developer demographics.

Key insights include:

- **JavaScript** remains the most widely used programming language.
- **Python** is gaining popularity and continues to grow in demand.
- **PostgreSQL and MongoDB** are becoming more popular database technologies.
- **Linux and Docker** dominate developer platform preferences.
- **React.js** shows strong future growth among web frameworks.
- The survey also highlights demographic patterns such as developer age groups, education levels, and geographic distribution.

These findings help developers and companies understand which technologies are becoming increasingly important.

INTRODUCTION

Stack Overflow conducts an annual survey to understand how developers work and which technologies they use.

The 2019 survey focuses on several areas:

- Programming languages
- Databases
- Platforms
- Web frameworks
- Developer demographics

The goal of this analysis is to identify current technology usage and future technology preferences. The results provide insights into how developer tools and technologies are evolving across the industry.

METHODOLOGY

The analysis followed several data processing and analytical steps.

Data Collection: Survey responses were collected from developers using an online questionnaire.

Data Cleaning: The dataset was cleaned by:

- Removing duplicate records
- Identifying and filling missing values
- Normalizing the dataset for analysis

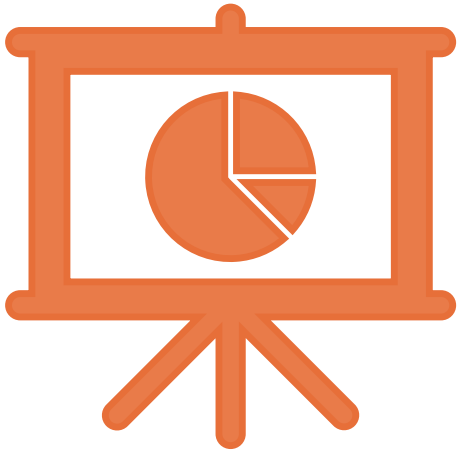
Data Analysis: Exploratory data analysis was conducted to identify patterns and trends.

Data Visualization: Several visualization methods were used, including:

- Bar charts
- Column charts
- Pie charts
- Maps
- Heat maps
- Word clouds

Tools such as Python, Excel, and Looker Studio dashboards were used to present the results.

Results



The analysis explores trends in multiple areas of developer technologies.

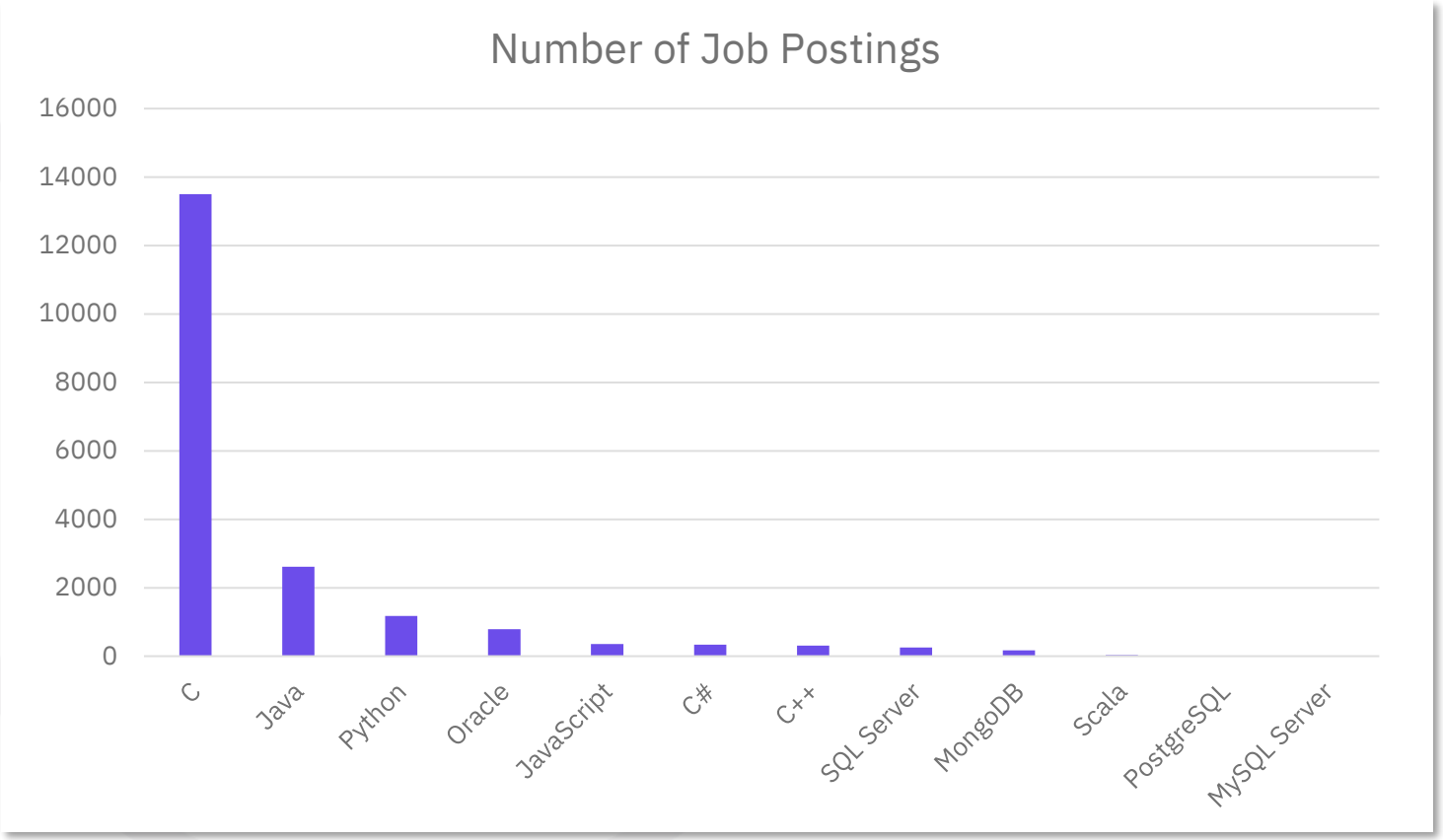
The results focus on:

- Programming language usage
- Database preferences
- Technology platforms
- Web frameworks
- Developer demographics

These results help identify **which technologies are dominant today and which technologies are gaining interest for the future**.

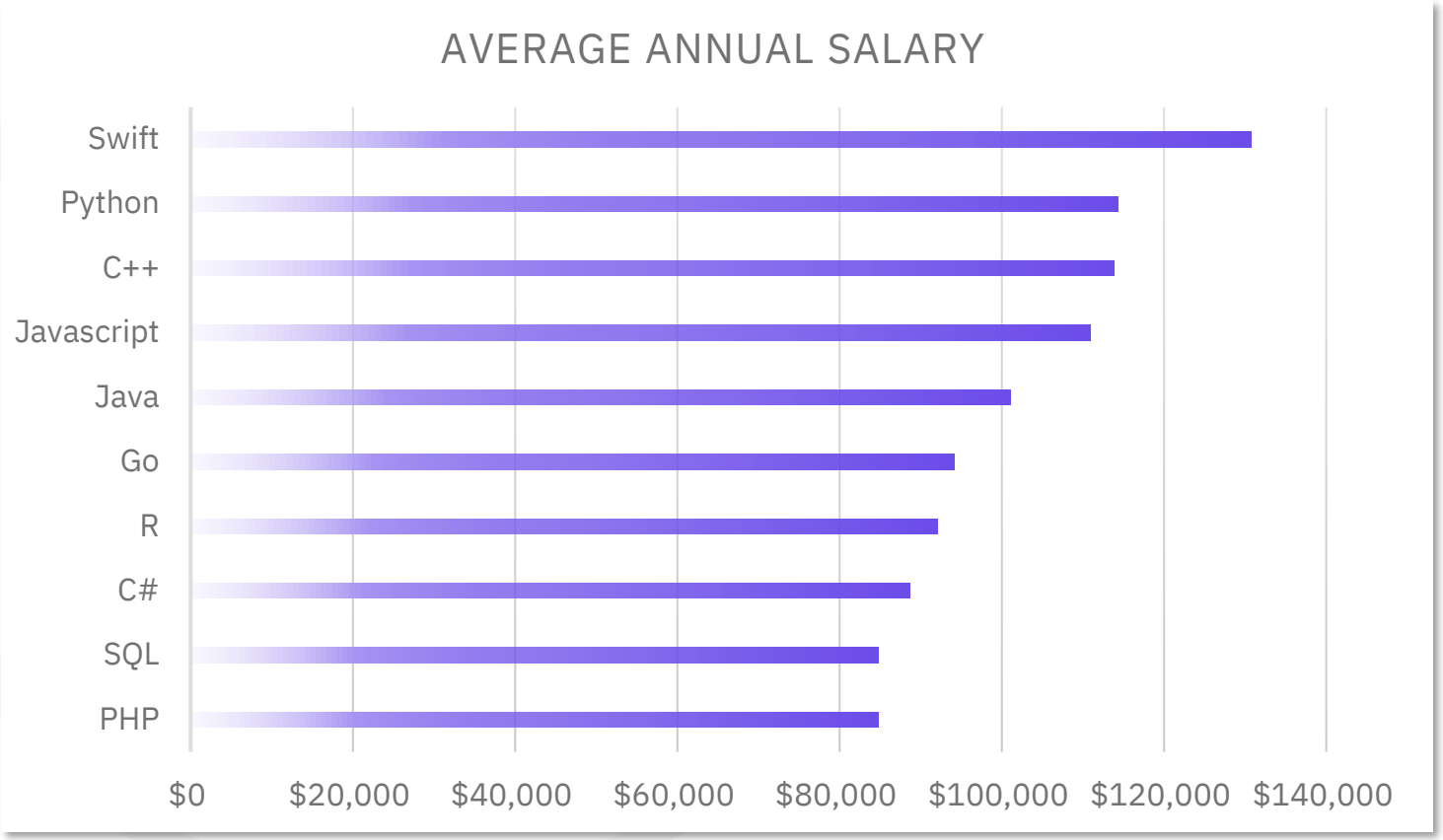
JOB POSTINGS

Module 1: Data collection using Job API



POPULAR LANGUAGES

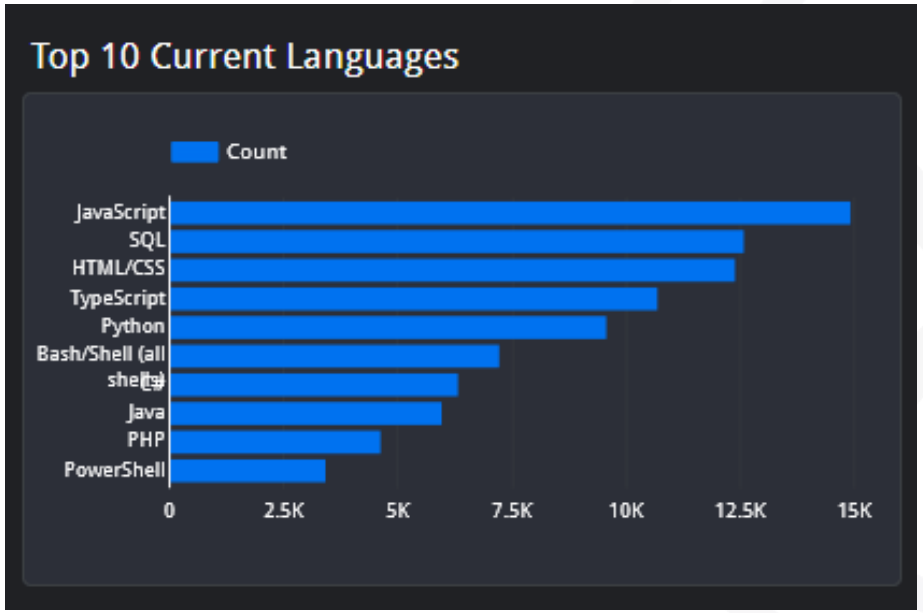
Module 1: Data collection using Web Scraping



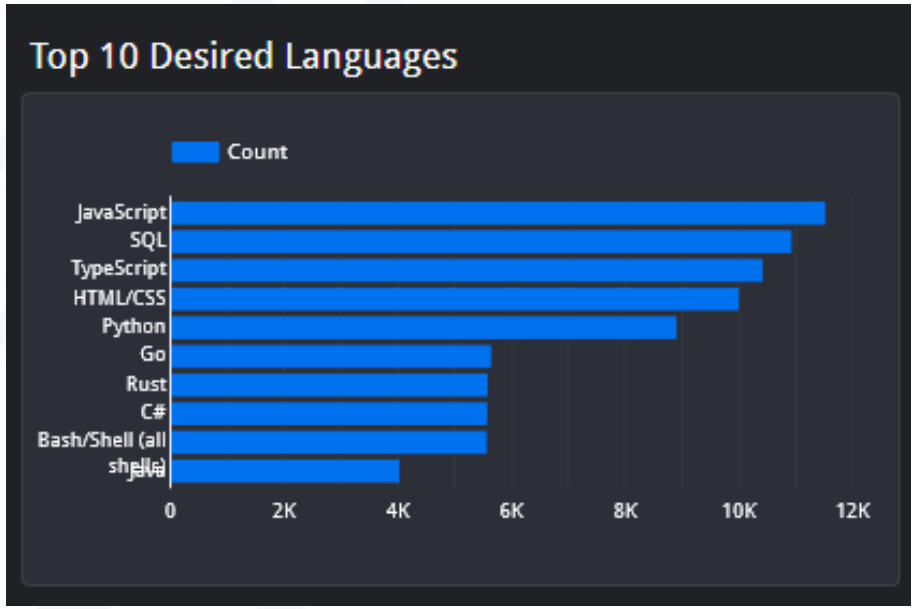
PROGRAMMING LANGUAGE TRENDS

Summaries of key trends shown in the charts.

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript remains the most widely used programming language.
- HTML/CSS and SQL continue to be essential technologies.
- Python shows strong growth and future demand.
- Go and TypeScript are becoming increasingly popular.

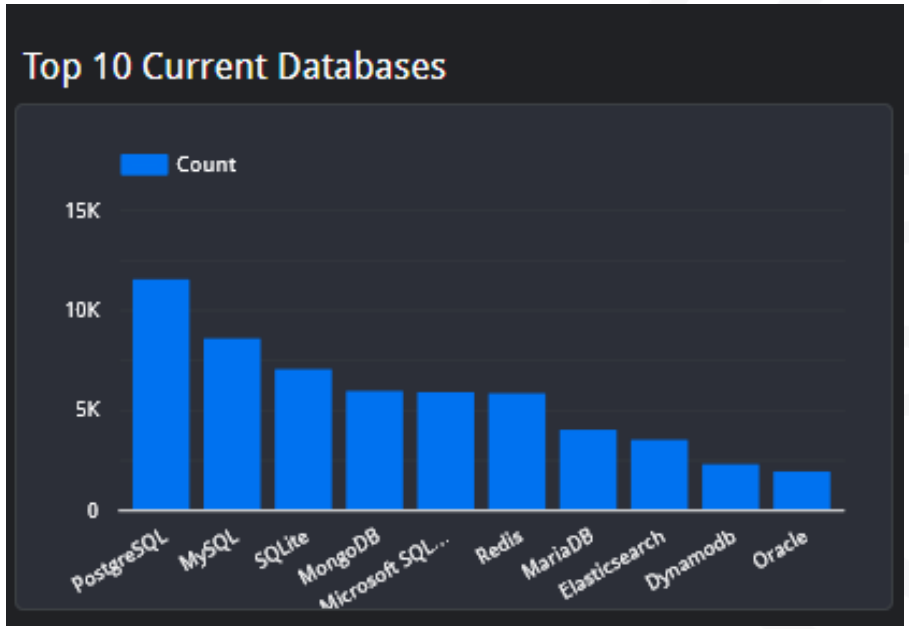
Implications

- Companies should prioritize hiring developers skilled in JavaScript and Python.
- SQL remains essential for data-related roles and database management.
- The growth of TypeScript and modern frameworks reflects the expansion of modern web development ecosystems.

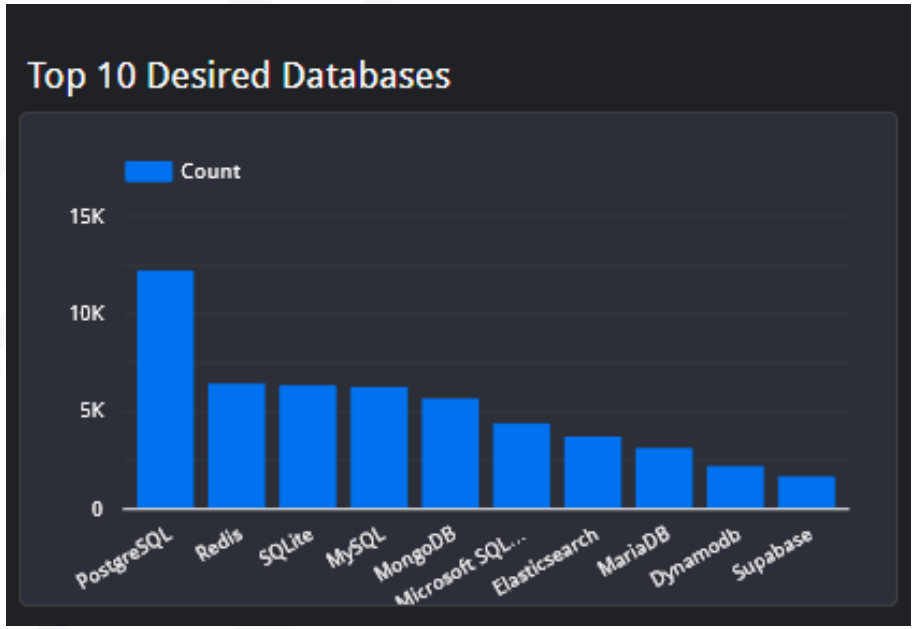
DATABASE TRENDS

Summaries of key trends shown in the charts.

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

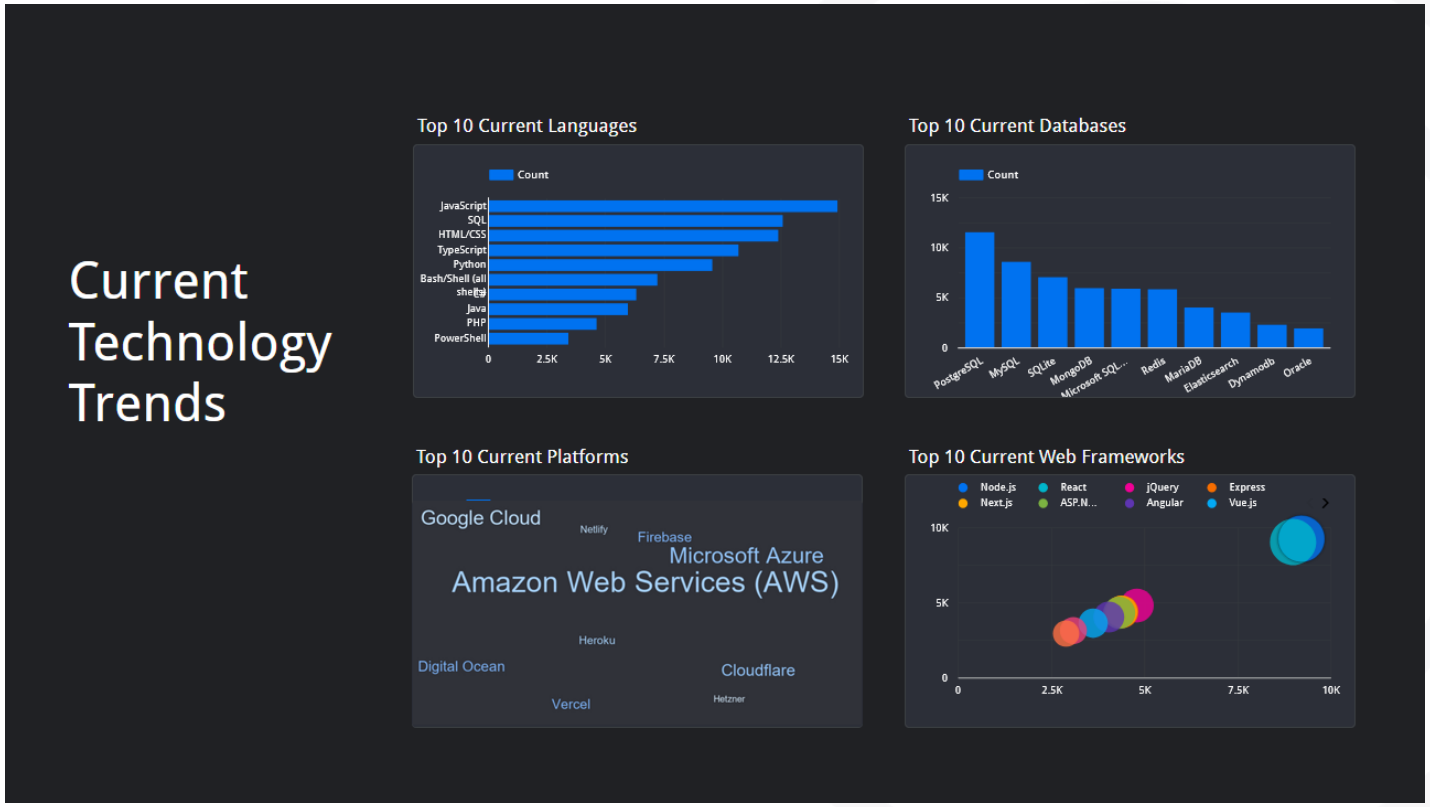
Findings

- MySQL remains widely used.
- PostgreSQL is one of the most desired databases for future projects.
- MongoDB is growing in popularity.
- Redis and Elasticsearch are emerging technologies.

Implications

- Companies are increasingly adopting open-source databases.
- PostgreSQL adoption continues to grow in modern applications.
- NoSQL databases play an important role in large-scale and flexible data systems.

DASHBOARD TAB 1: Current Technology Usage



Key Insights

- JavaScript, SQL, and HTML/CSS are the most widely used programming languages.
- PostgreSQL and MySQL lead among databases.
- AWS, Microsoft Azure, and Google Cloud dominate cloud platforms.
- React and Node.js are the most common web frameworks.

Summary

Web technologies, relational databases, and cloud platforms currently dominate the developer ecosystem.

DASHBOARD TAB 2: Future Technology Trends

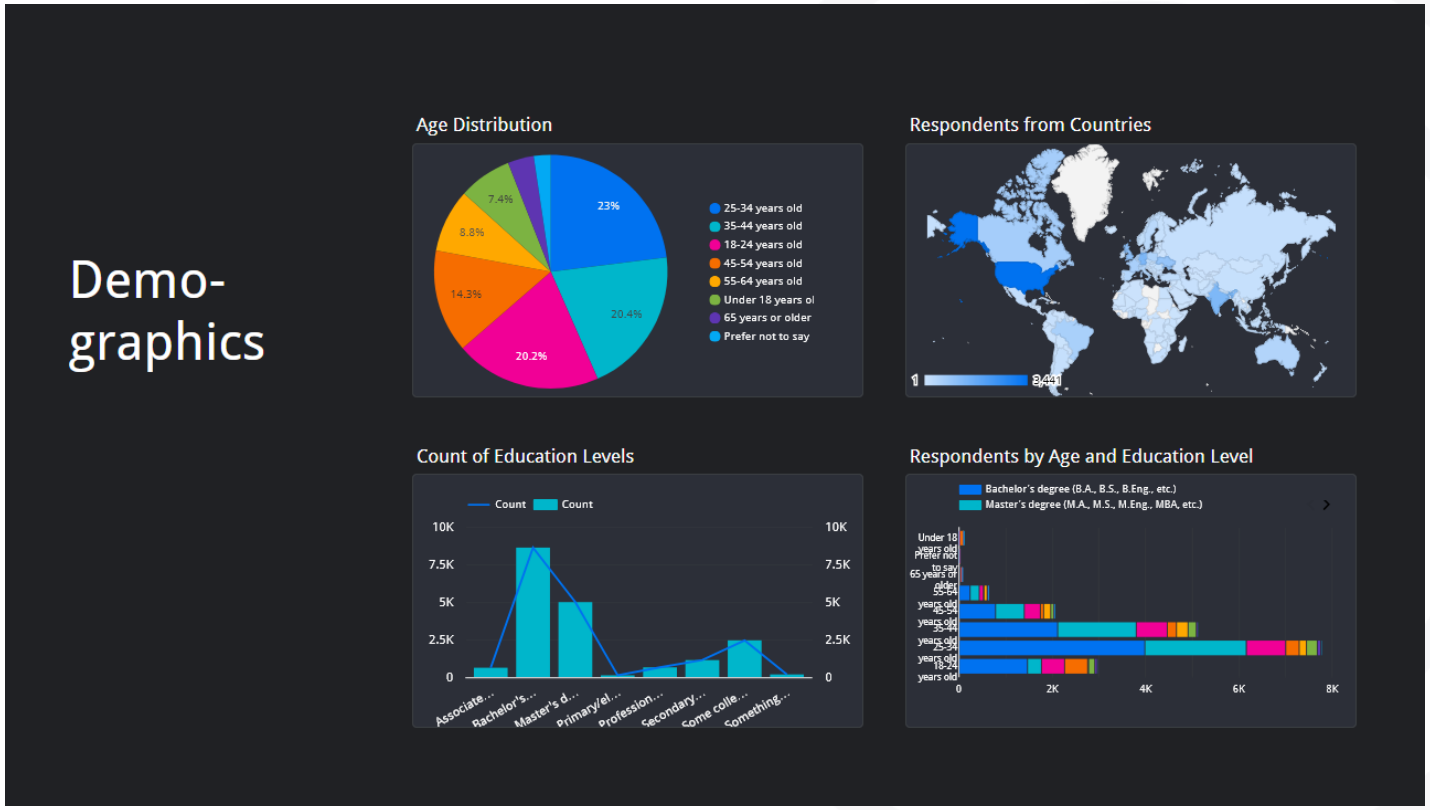
Key Insights

- JavaScript, SQL, and TypeScript remain the most desired languages.
- Go and Rust show growing interest among developers.
- PostgreSQL and Redis are among the most desired databases.
- AWS, Azure, and Google Cloud remain the most preferred platforms.

Summary

Future trends suggest continued growth in cloud computing, modern JavaScript frameworks, and emerging high-performance languages.

DASHBOARD TAB 3: Demographics



Key Insights

- Most respondents are 25–34 years old, followed by 35–44 and 18–24.
- Developers come from many countries worldwide.
- Most respondents hold Bachelor's or Master's degrees.

Summary

The developer population is young, globally distributed, and largely highly educated.

DISCUSSION

- The results show a strong dominance of web development technologies, especially JavaScript and related frameworks.
- Cloud platforms such as AWS, Azure, and Google Cloud play a central role in modern development environments.
- Developers continue to rely on relational databases like PostgreSQL and MySQL for data management.
- Interest in languages such as Go and Rust suggests growing demand for performance-oriented programming tools.
- The demographic results indicate that the developer workforce is young and globally distributed, with many holding higher education degrees.

OVERALL FINDINGS & IMPLICATIONS

Findings

- JavaScript and related web technologies remain the most widely used programming tools.
- Cloud platforms such as AWS, Microsoft Azure, and Google Cloud dominate the infrastructure landscape.
- PostgreSQL and MySQL continue to be the most commonly used databases.
- Emerging languages such as Go and Rust show growing interest among developers.
- The developer community is largely young and highly educated.

Implications

- Organizations should prioritize cloud computing and modern web technologies in their development strategies.
- Training programs should focus on modern frameworks, cloud platforms, and data technologies.
- Growing interest in new languages suggests a need to adapt to evolving programming ecosystems.
- Understanding developer demographics can help companies design better recruitment and skill development strategies.

CONCLUSION

- The analysis highlights clear patterns in technology adoption, developer preferences, and workforce demographics.
- Web technologies, cloud platforms, and relational databases remain central to modern software development.
- Future trends suggest increasing interest in modern programming languages and scalable infrastructure technologies.
- Understanding these trends can help organizations and developers adapt to the evolving technology landscape.